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THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA
Screening Test - Bhaskara Contest
(NMTC- at **JUNIOR LEVEL IX & X Standards**)
Saturday, 28th August 2010.

Note:

- 1) Fill in the response sheet with your Name, Class, the institution through which you appear in the specified places.
 - 2) Diagrams are only visual aids; they are not drawn to scale.
 - 3) You are free to do rough work on separate sheets.
 - 4) Duration of the test: 2 p.m. to 4. p.m. - 2 hours.
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PART - A

Note:

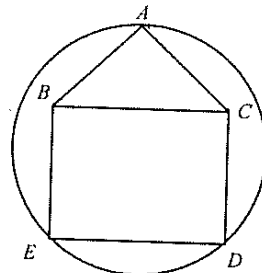
- Only one of the choices A,B,C,D is correct for each question. Shade that alphabet of your choice in the response sheet. (If you have any doubt in the method of answering, seek the guidance of your supervisor).
- For each correct response you get 1 mark; for each incorrect response you lose $\frac{1}{4}$ mark.

1. The number which, when subtracted from the terms of ratio $a:b$ makes it equal to $c:d$ is
(A) $\frac{ab - cd}{ab + cd}$ (B) $\frac{bc - ad}{c + d}$ (C) $\frac{ab + cd}{c + d}$ (D) $\frac{ab - cd}{b - c}$
2. In a Kilometre race Ram beats Shyam by 25 meters or 5 seconds. The time taken by Ram to complete the race is
(A) 1 minute (B) 5 minutes and 30 seconds
(C) 3 minutes and 15 seconds (D) 4 minutes and 10 seconds
3. Through D , the mid-point of the side BC of a triangle ABC , a straight line is drawn to meet AC at E and AB produced at F so that $AE=AF$. Then the ratio $BF:CE$ is
(A) 1:2 (B) 2:1 (C) 1:3 (D) None of these
4. In the bigger of two concentric circles two chords AB and AC are drawn to touch the smaller circle at D and E . Then BC is equal to
(A) $3DE$ (B) $4DE$ (C) $2DE$ (D) $\frac{3}{2}DE$
5. The number of solutions of the equation $x^{\log_{10} x} = 100x$ is
(A) 0 (B) 1 (C) 2 (D) 3

6. The internal bisector AE of the angle A of triangle is

(A) not greater than the median through A for all triangles.
 (B) not greater than the median through A for only acute angled triangles.
 (C) Not greater than the median through A for only obtuse angled triangles.
 (D) not less than the median through A for all triangles.

7. In the adjoining diagram ABC is an equilateral triangle and $BCDE$ is a square. The side of the equilateral triangle is 2010. The radius of the circle is



(A) 2010 (B) 4020 (C) 6030 (D) 8040

8. Given a and b are integers the expression $(a^2 + a + 2011)(2b + 1)$ is

(A) Odd for exactly 2010 values of a, b .
 (B) Odd for all values of a, b .
 (C) Even for exactly one value of a and two values of b
 (D) Odd for exactly for one value of a and one value of b .

9. A sequence of real number x_n is defined recursively as follows.

x_0, x_1 are arbitrary positive real numbers and $x_{n+2} = \frac{1 + x_{n+1}}{x_n}$ $n = 0, 1, 2, \dots$. Then the value of x_{2011} is

(A) 1 (B) x_0 (C) x_1 (D) x_2

10. If $xy = 6$ and $x^2y + y^2x + x + y = 63$, the value of $x^2 + y^2$ is

(A) 81 (B) 18 (C) 2010 (D) 78

11. If p is the perpendicular drawn from the vertex of a regular tetrahedron to the opposite face and if each edge is equal to 2 units. Then p is

(A) $8\sqrt{3}$ (B) $\frac{8\sqrt{3}}{2}$ (C) $\frac{8\sqrt{3}}{5}$ (D) $\frac{8\sqrt{3}}{3}$

12. The remainder when the polynomial $x + x^3 + x^9 + x^{27} + x^{81} + x^{243}$ is divided by $x^2 - 1$

(A) $6x$ (B) $2x$ (C) $3x$ (D) 1

13. Consider the sequence 4, 4, 8, 2, 0, 2, 2, 4, 6, 0, where the n^{th} term is the units place of the sum of the previous two terms for $n \geq 3$. If S_n is the sum to n terms of this sequence, then the smallest 'n' for which $s_n > 2010$ is
- (A) 253 (B) 502 (C) 503 (D) 504
14. P is a point inside an equilateral triangle of side 2010 units. The sum of the length of the perpendiculars drawn from P to the sides is equal to
- (A) 2010 (B) $2010\sqrt{3}$ (C) $1005\sqrt{3}$ (D) $\frac{2010}{\sqrt{3}}$
15. The equation $\log_{2x} \left(\frac{2}{x} \right) (\log_2 x)^2 + (\log_2 x)^4 = 1$ has
- (A) A root less than 1.
 (B) Has only one root greater than 1
 (C) Two irrational roots
 (D) No real roots

PART – B

Note:

- Write the correct answer in the space provided in the response sheet.
- For each correct response you get 1 mark; for each incorrect response you lose $\frac{1}{2}$ mark.

16. The value of $\sqrt[3]{20+14\sqrt{2}} + \sqrt[3]{20-14\sqrt{2}}$ is _____.
17. If a, b are positive and $a+b=1$ the minimum value of $a^4 + b^4$ is _____.
18. The whole surface area of rectangular block is 1332 cm^2 . The length, breadth and height are in the ratio 6:5:4. The sum of the length, breadth and height is _____ centimeters.
19. If $|x| + x + y = 10, x + |y| - y = 12$, then $x + y =$ _____.
20. Two parallel sides of a trapezoid are 3 and 9, the non parallel sides are 4 and 6. A line parallel to the bases (parallel sides) divides the trapezoid in to two trapezoids of equal perimeters. The ratio in which each of the non-parallel sides is divided as _____.
21. Triangle ABC has $AB = 17, AC = 25$ and the altitude to BC has length 15. The sum of the possible values of BC is _____.

22. $\frac{5}{6} = \frac{a_2}{2!} + \frac{a_3}{3!} + \frac{a_4}{4!} + \frac{a_5}{5!} + \frac{a_6}{6!}$ where $0 \leq a_i < i$, $i = 1, 2, 3, 4, 5, 6$. Then $a_2 + a_3 + a_4 + a_5 + a_6$ is _____.
23. A circle is circumscribed about a triangle with sides 30, 34, 16. It divides the circle into 4 regions with the non triangular regions being A, B, C ; C being the largest. Then the value of $(C - A - B)$ is _____.
24. If a number n is divisible by 8 and 30, then the smallest number of divisors that n has is _____.
25. Both the roots of the quadratic equation $x^2 - 12x + K = 0$ are prime numbers. The sum of all such values of K is _____.
26. In a convex polygon of 16 sides the maximum number of angles which can all be equal to 10° is _____.
27. If an arc of circle 1 subtending 60° at the centre, has double the length as the arc subtending 75° at the centre in circle 2, then $\frac{\text{area of circle 1}}{\text{area of circle 2}}$ is _____.
28. A two digit number is equal to the sum of the product of its digits and the sum of its digits. Then the units place of the number is _____.
29. Let $f(x)$ be a polynomial of degree 1. If $f(10) - f(5) = 15$, then $f(20) - f(5)$ equals _____.
30. The number of perfect square divisors of the number $12!$ is _____.